

Medical Insurance Claim Fraud Detection

AI-Driven Claim Verification & Explainable Fraud Detection Platform

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1. Problem Statement

Health insurance fraud — through falsified bills, exaggerated treatment costs, duplicate claims, or misrepresented medical records — leads to significant financial losses for insurers and higher premiums for genuine policyholders. Manual claim review is slow, inconsistent, and does not scale with claim volume. Equally important, when a claim is rejected or flagged, policyholders rarely receive a clear, understandable explanation of why. This project aims to build an AI-driven system that verifies claims end-to-end and, critically, explains the reasoning behind every decision.

2. Objectives

- Collect complete claim and policyholder information directly from the user — personal details, policy data, and supporting medical documents.
- Use OCR and Vision-Language Model (VLM) APIs to read, extract, and validate data from uploaded documents such as medical bills, prescriptions, and discharge summaries.
- Apply LLM-based agentic reasoning to cross-check claim details against policy rules, medical context, and historical claim data.
- Experiment with multiple modelling approaches for fraud classification and benchmark them to select the best-performing solution.
- Generate a clear, human-readable explanation for why a claim was approved, flagged, or rejected — not just a fraud/no-fraud label.

3. Proposed Approach

- **Data intake:** A guided user flow collects claimant, policy, and incident details along with document uploads (bills, prescriptions, reports, ID proofs).
- **Document intelligence:** OCR and VLM APIs extract structured fields (amounts, diagnoses, dates, provider details) from unstructured, often handwritten or scanned, medical documents.
- **Agentic AI pipeline:** Specialized agents handle document verification, policy-rule matching, anomaly/inconsistency detection, and final decision synthesis, coordinating rather than relying on a single monolithic model.
- **RAG (Retrieval-Augmented Generation):** Policy documents, claim history, and fraud rulebooks are retrieved and grounded into the LLM's reasoning so decisions are traceable to source evidence rather than hallucinated.

- **Model experimentation:** Multiple approaches — classical ML, deep learning, and LLM-based reasoning — are trialed on the fraud-classification task and compared on accuracy, precision/recall, and explainability to choose the strongest combination.
- **Explainable reasoning layer:** Every decision is paired with a natural-language justification, so a rejected or flagged claim comes with a specific, understandable reason.

4. Tech Stack

Layer	Technology
Application & Backend	Next.js with serverless functions (primary, for streamlined deployment & scaling); Node.js + Express as an alternative dedicated API backend
AI / Reasoning	LLM APIs, VLM APIs, Agentic AI orchestration, RAG pipeline
Document Processing	OCR pipeline for text extraction from scanned/uploaded medical documents
Data Layer	Relational DBMS for users, policies & claims; vector store for RAG retrieval
Deployment	Serverless hosting for scalable, low-maintenance deployment

5. Key Features

- End-to-end claim intake directly from the user, with no manual data re-entry.
- Automated document verification combining OCR and VLM understanding.
- Multi-agent fraud-detection pipeline grounded by RAG over policy and claim data.
- Transparent, explainable reasoning behind every approval, flag, or rejection.
- Multiple candidate approaches benchmarked to converge on the best-performing model.

6. Expected Outcome

A working prototype that takes a user's insurance claim and supporting documents, verifies authenticity through OCR/VLM-driven document intelligence, detects potential fraud through an agentic, RAG-grounded reasoning pipeline, and returns a transparent explanation for the outcome — reducing manual review effort and improving trust in the claims process for both insurers and policyholders.